

Markscheme

Mathematics: analysis and approaches

Practice question bank

23. (a)

Consider the function $f(x) = \frac{2x + 13}{(2x - 1)(x + 3)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{2x - 1} + \frac{B}{x + 3}$.

attempt to write $2x + 13 = A(x + 3) + B(2x - 1)$ (M1)

$2x + 13 = A(x + 3) + B(2x - 1)$ A1

substituting $x = \frac{1}{2}$: $14 = 3.5A \Rightarrow A = 4$ A1

substituting $x = -3$: $7 = -7B \Rightarrow B = -1$ A1

[4 marks]